

Message from the IEEE eScience 2023 Conference Leadership eScience 2023

The 19th IEEE Conference on eScience (eScience 2023), which took place from October 9th to 13th in Limassol (Cyprus), provided a platform where researchers, developers, and users of eScience applications and enabling IT technologies delved into the realms of interdisciplinary collaboration. The mission of the eScience conference was to drive innovation in data- and compute-intensive research, spanning a wide array of disciplines, including the physical and biological sciences, as well as the social sciences, arts, and humanities. IEEE eScience 2023 successfully dismantled traditional barriers, nurturing collaboration among interdisciplinary research communities, developers, and eScience application users.

Throughout that five-day event, the conference focus was on advancing all aspects of eScience and its associated technologies, applications, algorithms, and tools, with a strong emphasis on practical solutions to real-world challenges.

New Directions and Communities: This year, we have introduced four key topics: "Computational Science for sustainable development," "FAIR," "Research Infrastructures for eScience," and "Continuum Computing: Convergence between Cloud Computing and the Internet of Things (IoT)." These tracks enriched our discussions and investigations.

Keynote Speakers: In addition to these exciting tracks, we were delighted to present our distinguished keynote speakers: Dr. İlkyay Altıntaş illuminated the convergence of machine learning, AI, and scientific research, showcasing innovative approaches and real-world applications. Professor Ian T. Foster took us on a journey into the realm of global science services, demonstrating how they can reshape collaborative research. Finally, Professor Paul Watson shared insights from the National Innovation Centre for Data, highlighting successful projects that leverage data science and AI to drive impact.

Program Overview: As we embarked on this eScience journey, we invited participants to engage, collaborate, and explore the transformative potential of eScience and its associated technologies. Together, we addressed practical solutions, open challenges, and pushed the boundaries of interdisciplinary research. This year's conference featured three keynotes from different domains, presented 40 peer-reviewed papers, showcased 23 posters, featured 15 invited talks from renowned researchers, hosted 6 workshops, and provided 5 tutorials. The conference was held in person and was co-located with the 4th Global Research Platform Workshop (4GRP).

Our sincere appreciation goes out to the authors who submitted exceptional papers, and we are immensely thankful for the dedicated efforts of our numerous volunteers. The Technical Paper Committee diligently reviewed 96 papers and 25 posters, with over 70 volunteers conducting 308 reviews. This collaborative endeavour has resulted in the exceptional collection of papers featured in this volume.

We would also like to express our deep gratitude to the IEEE Computer Society and IEEE's Technical Committee on High-Performance Computing (TCHPC) for their unwavering sponsorship and support, which have been instrumental in making this conference possible.

Finally, we extend our gratitude to you, our valued readers, for your interest in this volume. We are confident that the contents within will greatly contribute to the advancement of your research, whether it pertains to eScience or other domains.

Warm regards,

George A. Papadopoulos
IEEE eScience 2023 General Chairs

Rafael Ferreira da Silva
Rosa Filgueira
IEEE eScience 2023 Technical Program Chairs